

$^{27}\text{Al}(^{12}\text{C},\alpha n\gamma)$ 2014Bi03

2014Bi03: a 40-MeV ^{12}C beam was provided from the 14-UD Pelletron accelerator at the Tata Institute of Fundamental Research, Mumbai. The target was 0.50-mg/cm² ^{27}Al on 10-mg/cm² Au backing. States of ^{34}Cl were populated via the $^{27}\text{Al}(^{12}\text{C},\alpha n)^{34}\text{Cl}$ heavy-ion reaction and the deexciting γ rays were detected using the Indian National Gamma Array, consisting of 15 Compton-suppressed clover detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (DCO), and integrated polarization asymmetry (IPDCO). Deduced levels, J, π , γ -ray multipolarities, lifetimes from DSAM lineshape analysis, and transition strengths. Comparisons with large-basis shell-model calculations.

1978Ba39: 27-45-MeV ^{12}C beams from the Strasbourg tandem accelerator, France, impinged on 350- $\mu\text{g}/\text{cm}^2$ Al targets on Ta backings. γ rays were detected using two Ge(Li) detectors. Measured $\gamma\gamma$ -coin and $\gamma\gamma(\theta)$. Deduced levels, J, π , γ -ray multipolarities, and lifetimes using the Recoil Distance Method (RDM). Also measured the $^{31}\text{P}(\alpha,n\gamma)^{34}\text{Cl}$ reaction.

 ^{34}Cl Levels

E(level) [†]	J π [‡]	T _{1/2}	Comments
0.0	0 ⁺		
146.36 3	3 ⁺		
460.98 26	1 ⁺		
666.2 5	1 ⁺		
1230.2 5	2 ⁺		
1887.2 6	2 ⁺		
2158.2 5	2 ⁺		
2181.3 5	3 ⁺		
2375.8 5	4 ⁺		
2611.3 5	3 ⁺		
2721.13 34	2 ⁻		
3600.46 32	4 ⁻		
3631.7 4	5 ⁻		
3646.5 7	5 ⁺		
4371.5 13	4 ⁺	>2.08 ps	T _{1/2} : $\tau > 3$ ps (1978Ba39, RDM).
4743.59 34	6 ⁻		
4824.52 34	5 ⁺		
4862.4 8	5 ⁻		
5315.4 4	7 ⁺	65 ps 8	T _{1/2} : $\tau = 94$ ps 12 (1978Ba39, RDM).
7250.5 11	9 ⁺	0.159 ps 28	T _{1/2} : $\tau = 0.23$ 4 ps (2014Bi03, DSAM).
7699.5 11	(7 ⁺)	<0.34 ps	T _{1/2} : $\tau < 0.49$ ps (2014Bi03, DSAM).
7800.5 11	8 ⁻	<83.0 fs	T _{1/2} : $\tau < 0.12$ ps (2014Bi03, DSAM).
8155.5 11	6 ⁺	<56 fs	T _{1/2} : $\tau < 0.08$ ps (2014Bi03, DSAM).
9392.7 11	9 ⁻	<6.9 fs	T _{1/2} : $\tau < 0.01$ ps (2014Bi03, DSAM).
10631.7 15	11 ⁺	<28 fs	T _{1/2} : $\tau < 0.04$ ps (2014Bi03, DSAM).

[†] From a least-squares fit to γ -ray energies.

[‡] For g.s. and the 146 level, J^π is adopted from the Adopted Levels; for higher levels, J^π is assigned in 2014Bi03 based on their γ -ray multipolarities.

γ(³⁴Cl)

<u>E_γ[†]</u>	<u>I_γ[†]</u>	<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>δ</u>	<u>γ branching ratio[#]</u>	<u>Comments</u>
146.36 3		146.36	3 ⁺	0.0	0 ⁺	[M3]			E _γ : from the Adopted Levels.
453 1	7.9 4	2611.3	3 ⁺	2158.2	2 ⁺	D+Q	+0.11 6	19 13	R _{DCO} (D)=1.30 17, 3 ⁺ to 2 ⁺ from 2014Bi03 level scheme.
453 1		5315.4	7 ⁺	4862.4	5 ⁻			2.8 2	
460.9 3	100 7	460.98	1 ⁺	0.0	0 ⁺	M1			E _γ : weighted average of 460.9 3 (1978Ba39) and 461 1 (2014Bi03).
									γ branching ratio: other: 100 (1978Ba39).
									R _{DCO} (Q)=0.49 2, POL=-0.08 1, 1 ⁺ to 0 ⁺ from 2014Bi03 level scheme.
									A ₂ =-0.25 4 (1978Ba39).
490.9 3	253 13	5315.4	7 ⁺	4824.52	5 ⁺	E2		72 4	E _γ : weighted average of 490.9 3 (1978Ba39) and 491 1 (2014Bi03).
									δ: 0.00 5 (1978Ba39).
									γ branching ratio: other: 66 2 (1978Ba39).
									R _{DCO} (Q)=0.99 3, POL=+0.14 2, 7 ⁺ to 5 ⁺ from 2014Bi03 level scheme.
									A ₂ =0.14 2, A ₄ =-0.08 2 (1978Ba39).
563 1	12.4 7	2721.13	2 ⁻	2158.2	2 ⁺	D+Q	-0.38 6	9.8 11	R _{DCO} (D)=1.13 10, 2 ⁻ to 2 ⁺ from 2014Bi03 level scheme.
564 1		1230.2	2 ⁺	666.2	1 ⁺			36 5	
571.8 3	21.8 11	5315.4	7 ⁺	4743.59	6 ⁻	E1+M2	-0.04 2	21.6 11	E _γ : weighted average of 571.8 3 (1978Ba39) and 572 1 (2014Bi03).
									δ: other: -0.03 12 (1978Ba39).
									γ branching ratio: other: 34 2 (1978Ba39).
									R _{DCO} (Q)=0.44 3, POL=+0.15 10, 7 ⁺ to 6 ⁻ from 2014Bi03 level scheme.
									A ₂ =-0.08 4 (1978Ba39).
									Note that the branching ratio of 21.6 2 in Table II of 2014Bi03 differs from 7.35 7 deduced from the relative intensity in Table I of 2014Bi03.
666 1	45.0 23	666.2	1 ⁺	0.0	0 ⁺	M1			R _{DCO} (Q)=0.48 4, POL=-0.10 3, 1 ⁺ to 0 ⁺ from 2014Bi03 level scheme.
725 1	5.0 3	4371.5	4 ⁺	3646.5	5 ⁺	M1+E2			R _{DCO} (Q)=0.38 7, POL=-0.17 12, 4 ⁺ to 5 ⁺ from 2014Bi03 level scheme.
769 1	19.4 10	1230.2	2 ⁺	460.98	1 ⁺	D+Q	+0.04 3	32 5	R _{DCO} (D)=1.13 9, 2 ⁺ to 1 ⁺ from 2014Bi03 level scheme.
834 1		2721.13	2 ⁻	1887.2	2 ⁺			2.0 2	
879.4 4	98 5	3600.46	4 ⁻	2721.13	2 ⁻	E2		49.2 31	E _γ : weighted average of 879.5 5 (1978Ba39) and 879 1 (2014Bi03).
									R _{DCO} (Q)=0.95 5, POL=+0.12 2, 4 ⁻ to 2 ⁻ from 2014Bi03 level scheme.
									A ₂ =0.19 8 (1978Ba39).
928 1		2158.2	2 ⁺	1230.2	2 ⁺			8.6 11	
1084 1		1230.2	2 ⁺	146.36	3 ⁺			31 4	
1111.8 6	29.5 15	4743.59	6 ⁻	3631.7	5 ⁻	M1+E2	+0.36 6	29.2 32	E _γ : weighted average of 1111.6 8 (1978Ba39) and 1112 1

$^{27}\text{Al}(^{12}\text{C}, \alpha n \gamma)$ **2014Bi03 (continued)**

$\gamma(^{34}\text{Cl})$ (continued)									Comments
$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ	γ branching ratio [#]	
1143.0 6	34.4 18	4743.59	6 ⁻	3600.46	4 ⁻	E2		34 4	(2014Bi03). γ branching ratio: other: 30 2 (1978Ba39). $R_{\text{DCO}}(\text{D})=1.92$ 17, $\text{POL}=-0.03$ 1, 6 ⁻ to 5 ⁻ from 2014Bi03 level scheme. $A_2=-0.08$ 20 (1978Ba39). E_γ : weighted average of 1143.0 8 (1978Ba39) and 1143 1 (2014Bi03). δ : 0.09 53 (1978Ba39). γ branching ratio: other: 43 2 (1978Ba39). $R_{\text{DCO}}(\text{D})=2.08$ 18, $\text{POL}=+0.07$ 2, 6 ⁻ to 4 ⁻ from 2014Bi03 level scheme. $A_2=0.16$ 9, $A_4=-0.16$ 9 (1978Ba39). $R_{\text{DCO}}(\text{D})=1.94$ 36, consistent with $\Delta J=0$, $\text{POL}=+0.08$ 2, 5 ⁺ to 5 ⁻ from 2014Bi03 level scheme. E_γ : weighted average of 1223.8 8 (1978Ba39) and 1224 1 (2014Bi03). γ branching ratio: other: 33 5 (1978Ba39). $R_{\text{DCO}}(\text{Q})=0.42$ 2, $\text{POL}=+0.04$ 1, 5 ⁺ to 4 ⁻ from 2014Bi03 level scheme. $A_2=-0.50$ 6 (1978Ba39). E_γ : weighted average of 1223.8 8 (1978Ba39) and 1225 1 (2014Bi03). $R_{\text{DCO}}(\text{Q})=1.29$ 14, consistent with $\Delta J=0$, 4 ⁻ to 4 ⁺ from 2014Bi03 level scheme. E_γ : weighted average of 1256.0 15 (1978Ba39) and 1256 1 (2014Bi03). γ branching ratio: other: 67 2 (1978Ba39). $R_{\text{DCO}}(\text{Q})=0.39$ 5, $\text{POL}=+0.010$ 2, 5 ⁻ to 4 ⁺ from 2014Bi03 level scheme. $A_2=-0.29$ 7 (1978Ba39). $R_{\text{DCO}}(\text{D})=0.71$ 11, 2 ⁺ to 1 ⁺ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{D})=2.24$ 20, no POL data but E2 is assigned by 2014Bi03, 3 ⁺ to 1 ⁺ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{Q})=1.03$ 38 from 65°–90° matrix. 2014Bi03 assigns E2. 7 ⁺ to 5 ⁺ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{D})=0.74$ 6, 2 ⁺ to 1 ⁺ from 2014Bi03 level scheme. $R_{\text{DCO}}(\text{Q})=1.10$ 16, $\text{POL}=+0.15$ 6, 3 ⁺ to 1 ⁺ from 2014Bi03 level scheme.
1193 1	8.9 5	4824.52	5 ⁺	3631.7	5 ⁻	E1(+M2)	-0.1 3	5.1 3	
1223.9 6	34.7 18	4824.52	5 ⁺	3600.46	4 ⁻	E1(+M2)	-0.02 2	14.8 8	
1224.3 6	15.5 8	3600.46	4 ⁻	2375.8	4 ⁺	D(+Q)	+0.3 3	7.8 6	
1256.0 8	23.0 12	3631.7	5 ⁻	2375.8	4 ⁺	E1(+M2)	-0.01 4	52.5 29	
1381 1		2611.3	3 ⁺	1230.2	2 ⁺			28.0 25	
1426 1	8.5 5	1887.2	2 ⁺	460.98	1 ⁺	D+Q	-0.11 5	48 4	
1491 1		2721.13	2 ⁻	1230.2	2 ⁺			2.1 2	
1515 1	14.5 7	2181.3	3 ⁺	666.2	1 ⁺	Q		14.5 19	
1669 1	13.8 8	5315.4	7 ⁺	3646.5	5 ⁺	Q		3.5 3	
1697 1	22.3 11	2158.2	2 ⁺	460.98	1 ⁺	D+Q	-0.10 2	62 4	
1720 1	41.3 21	2181.3	3 ⁺	460.98	1 ⁺	E2		41 5	
1741 1		1887.2	2 ⁺	146.36	3 ⁺			52 5	

²⁷Al(¹²C,αnγ) **2014Bi03** (continued)

^γ(³⁴Cl) (continued)

E_γ [†]	I_γ [†]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ	γ branching ratio [#]	Comments
1935 1	232 12	7250.5	9 ⁺	5315.4	7 ⁺	E2			R _{DCO} (Q)=1.10 3, POL=+0.08 1, 9 ⁺ to 7 ⁺ from 2014Bi03 level scheme.
1945 1		2611.3	3 ⁺	666.2	1 ⁺			17.0 16	
2012 1		2158.2	2 ⁺	146.36	3 ⁺			10.2 11	
2035 1	2.9 2	2181.3	3 ⁺	146.36	3 ⁺	E1+M2	-0.4 1	44 6	R _{DCO} (Q)=0.51 8, POL=+0.04 1, 3 ⁺ to 3 ⁺ from 2014Bi03 level scheme, B(M1) is calculated in 2014Bi03 Table V. This is a discrepancy. Note that the branching ratio of 44.1 46 in Table II of 2014Bi03 differs from 4.9 2 deduced from the relative intensity in Table I of 2014Bi03 .
2055 1	13.5 7	2721.13	2 ⁻	666.2	1 ⁺	D+Q	-0.05 3	7.5 9	R _{DCO} (D)=0.87 6, 2 ⁻ to 1 ⁺ from 2014Bi03 level scheme.
2158 1		2158.2	2 ⁺	0.0	0 ⁺			19.1 22	
2213 1	2.1 2	4824.52	5 ⁺	2611.3	3 ⁺			1.6 1	R _{DCO} (Q)=0.69 15. No multipolarity or δ is given in 2014Bi03 . 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
2229 1	40.9 21	2375.8	4 ⁺	146.36	3 ⁺	M1+E2	+0.4 1		R _{DCO} (Q)=0.90 9, POL=-0.09 3, 4 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
2260 1	88 5	2721.13	2 ⁻	460.98	1 ⁺	E1+M2	+0.05 2	49 6	R _{DCO} (D)=1.15 5, POL=+0.04 1, 2 ⁻ to 1 ⁺ from 2014Bi03 level scheme.
2384 1	18.2 9	7699.5	(7 ⁺)	5315.4	7 ⁺	M1			R _{DCO} (Q)=0.89 25 from 65°-90° matrix, consistent with $\Delta J=0$, POL=-0.17 11. No multipolarity or δ is given in 2014Bi03 . B(M1) is calculated in 2014Bi03 Table V. (7 ⁺) to 7 ⁺ from 2014Bi03 level scheme.
2449 1		4824.52	5 ⁺	2375.8	4 ⁺			1.5 1	
2465 1		2611.3	3 ⁺	146.36	3 ⁺			36.0 31	
2485 1	89 5	7800.5	8 ⁻	5315.4	7 ⁺	E1+M2	+0.22 7		R _{DCO} (Q)=0.74 8, POL=+0.016 2, 8 ⁻ to 7 ⁺ from 2014Bi03 level scheme. B(M2)(W.u.)>61.1 exceeds RUL=3.
2562 1		4743.59	6 ⁻	2181.3	3 ⁺	[E3]		1.0 1	
2575 1	30.2 16	2721.13	2 ⁻	146.36	3 ⁺	E1+M2	-0.3 1	16.8 19	R _{DCO} (Q)=0.51 12, POL=+0.03 1, 2 ⁻ to 3 ⁺ from 2014Bi03 level scheme.
2643 1	14.7 8	4824.52	5 ⁺	2181.3	3 ⁺	Q		4.8 7	R _{DCO} (Q)=1.13 13. 2014Bi03 assigns E2. 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme.
2681 1	3.3 2	4862.4	5 ⁻	2181.3	3 ⁺	M2			R _{DCO} (D)=1.88 26, POL=-0.1 1. No multipolarity or δ is given in 2014Bi03 . 5 ⁻ to 3 ⁺ from 2014Bi03 level scheme.
2721 1	23.2 12	2721.13	2 ⁻	0.0	0 ⁺	M2		12.9 15	R _{DCO} (Q)=0.98 29, POL=-0.06 3, 2 ⁻ to 0 ⁺ from 2014Bi03 level scheme.
2840 1	47.3 24	8155.5	6 ⁺	5315.4	7 ⁺	M1+E2	-0.3 1		R _{DCO} (Q)=0.83 12, POL=-0.02 1, 6 ⁺ to 7 ⁺ from 2014Bi03 level scheme.
3381 1	74 4	10631.7	11 ⁺	7250.5	9 ⁺	E2			R _{DCO} (Q)=1.00 10, POL=+0.03 2, 11 ⁺ to 9 ⁺ from 2014Bi03 level scheme.
3453.8 6	85 4	3600.46	4 ⁻	146.36	3 ⁺	M1(+E2)	+0.05 5	43.0 28	E _γ : weighted average of 3453.6 8 (1978Ba39) and 3454 1 (2014Bi03). R _{DCO} (Q)=0.68 5, POL=-0.008 1, 4 ⁻ to 3 ⁺ from 2014Bi03

²⁷Al(¹²C,αnγ) **2014Bi03** (continued)

γ(³⁴Cl) (continued)

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	δ	γ branching ratio [#]	Comments
3485.1 6	15.4 8	3631.7	5 ⁻	146.36	3 ⁺	M2+E3	-0.23 12	47.5 27	level scheme. This is a discrepancy. A ₂ =-0.22 8, A ₄ =-0.11 8 (1978Ba39). E _γ : weighted average of 3485.2 8 (1978Ba39) and 3485 1 (2014Bi03). γ branching ratio: other: 33 2 (1978Ba39). R _{DCO} (Q)=1.74 47, POL=-0.10 4, 5 ⁻ to 3 ⁺ from 2014Bi03 level scheme. A ₂ =0.04 14 (1978Ba39). R _{DCO} (Q)=0.88 11, POL=+0.003 3, 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme. R _{DCO} (Q)=1.20 43, POL=-0.02 1, 9 ⁻ to 7 ⁺ from 2014Bi03 level scheme. B(M2)(W.u.)>1208 exceeds RUL=3. B(E3)(W.u.)<7.57×10 ¹¹ upper limit exceeds RUL=100. E _γ : weighted average of 4597.1 10 (1978Ba39) and 4597 1 (2014Bi03). γ branching ratio: other: 27 3 (1978Ba39). R _{DCO} (D)=3.45 26, POL=+0.010 2, 6 ⁻ to 3 ⁺ from 2014Bi03 level scheme. E _γ : weighted average of 4677.7 10 (1978Ba39) and 4678 1 (2014Bi03). δ: -0.03 10 (1978Ba39). γ branching ratio: other: 67 5 (1978Ba39). R _{DCO} (Q)=1.00 2, POL=+0.010 1, 5 ⁺ to 3 ⁺ from 2014Bi03 level scheme. A ₂ =0.19 4, A ₄ =-0.09 4 (1978Ba39).
3500 1	37.2 19	3646.5	5 ⁺	146.36	3 ⁺	E2			
4077 1	17.9 10	9392.7	9 ⁻	5315.4	7 ⁺	M2(+E3)	+0.2 5		
4597.1 7	36.2 19	4743.59	6 ⁻	146.36	3 ⁺	(E3)		36 5	
4677.9 7	173 9	4824.52	5 ⁺	146.36	3 ⁺	E2		72 4	

[†] From 2014Bi03, unless otherwise noted. No uncertainties for E_γ are provided in 2014Bi03. Based on the typical uncertainties reported in other work by the authors using the same INGA detector array, the evaluators assume a ΔE_γ=1 keV for averaging and least-squares fitting. 2014Bi03 Table I provides only statistical uncertainties for I_γ. The systematic uncertainties are ≈5%, arising from the efficiency curve and fits, as communicated by M. Saha Sarkar in an email reply to B. Singh on February 17, 2014. The evaluators have combined the statistical and systematic uncertainties in quadrature.

[‡] Deduced by evaluators based on measured γγ(θ)(DCO) and γγ(θ)(POL) in 2014Bi03. R_{DCO}(D) is from gating on a known stretched dipole (ΔJ=1) transition and R_{DCO}(Q) is from gating on a known stretched quadrupole (ΔJ=2) transition. For stretched transition with same multipolarity as the gating transition, R_{DCO} value should be very close to unity. For different multiplicities of the gating and projected transitions, the R_{DCO} value depends on the angle between the detectors and the amount of mixing present in the mixed multipolarity transition. However, the authors did not specify the expected R_{DCO} values. Based on Fig. 5 in 2014Bi03, the expected R_{DCO}(Q)≈1.0 and R_{DCO}(D)≈2.0 for stretched quadrupole (ΔJ=2) or unstretched dipole (ΔJ=0) transitions; R_{DCO}(Q)≈0.5 and R_{DCO}(D)≈1.0 for stretched dipole (ΔJ=1) transitions. A positive POL indicates an electric character, and a negative POL indicates a magnetic character. Twelve multiplicities are assigned in 1978Ba39, which are consistent with the 2014Bi03 assignments.

[#] 2014Bi03 Table II provides only statistical uncertainties for the γ branching ratios. According to the email reply to B. Singh from the corresponding author M. Saha Sarkar on February 17, 2014, the γ branching ratios listed in Table II of 2014Bi03 were measured separately by gating on transitions above the level. The systematic

$\gamma(^{34}\text{Cl})$ (continued)

uncertainties for these measurements are $\approx 5\%$, which arise from the efficiency curve and fits. The evaluators have combined the statistical and systematic uncertainties in quadrature. In some cases, the branching ratios differ significantly from those derived from the I γ data in Table I of [2014Bi03](#). The I γ data in Table I should be considered more reliable, as the authors suggested in their email reply.

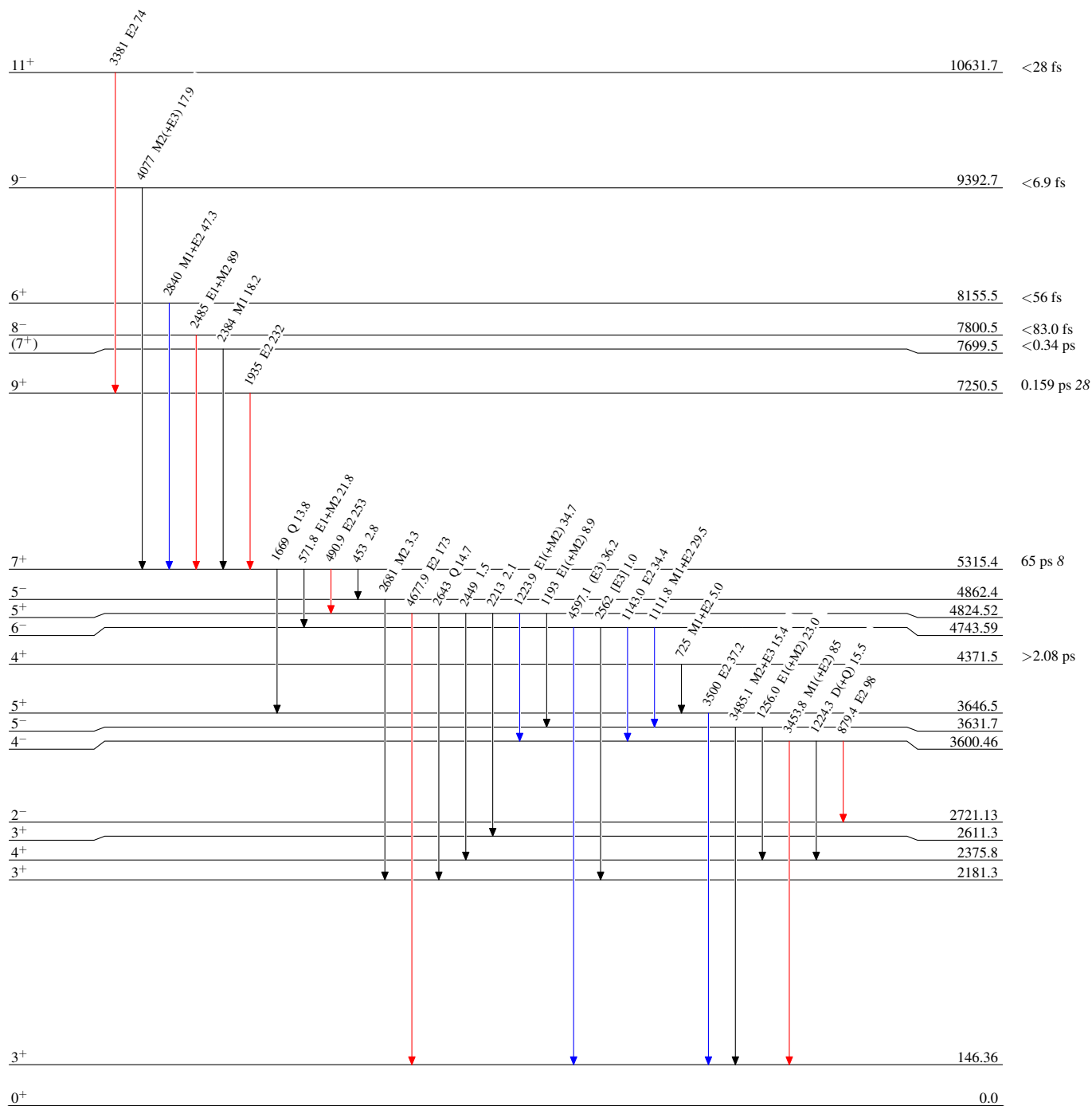
$^{27}\text{Al}(^{12}\text{C}, \alpha n \gamma)$ 2014Bi03

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



²⁷Al(¹²C,any) **2014B103**

Level Scheme (continued)

Intensities: Relative I_γ

